



Learning Linux

Creating a Fedora Remix

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Fedora Remix Builder - Container Quickstart Guide

Quick guide to building a custom Fedora Remix ISO using the Fedora Remix Builder container ([Build_Remix.sh](#) / [Verify_Build_Remix.sh](#)).

1. Overview

This guide walks you through a **Podman-based** remix build: the workload runs inside the **fedora-remix-builder** image; your host stays outside the compose except for bind-mounted workspace and ISO output paths.

Standard configuration order (recommended)

1. **Clone:** `git clone https://github.com/tmichett/Fedora_Remix.git` and `cd Fedora_Remix`.
2. **Configure:** run `./Update_Remix_Config.sh` — prompts for `SSH_Key_Location`, `Fedora_Remix_Location`, `GitHub_Registry_Owner`, `Fedora_Version`, and PXE (`include_pxeboot_files`). Press **Enter** at any prompt to keep the value shown in brackets (defaults come from `config.yml`). Optionally pre-seed `config.yml` so those defaults match your host (see **Step 2**).
3. **Verify / build:** `./Verify_Build_Remix.sh` (recommended) or `./Build_Remix.sh`.



If you use the **published** builder image at `ghcr.io/tmichett/fedora-remix-builder`, keep `GitHub_Registry_Owner` set to `tmichett` so pulls and verification resolve the same image (the script reminds you before that prompt).

PXE/web assets (`Prepare_Web_Files.py` inside the build): enable `include_pxeboot_files` only when you intentionally stage `vmlinux` / `initrd` for network boot tooling. Use **false** (answer **no**) for a normal ISO-only build—see IMPORTANT under **Step 3** below; fragile mirrors on older or unreleased Fedora can fail the prepare step if PXE downloads are forced on.

Build Time	Approximately 30-45 minutes
Output	A bootable Fedora Remix ISO file (~7-8 GB)

2. Prerequisites

2.1. Required Software

Before starting, ensure your system has:

1. **Git** - For cloning the repository

```
# Fedora/RHEL
sudo dnf install git

# Ubuntu/Debian
sudo apt install git
```

2. **Podman** - Container runtime (recommended) or Docker

```
# Fedora/RHEL
sudo dnf install podman

# Ubuntu/Debian
sudo apt install podman
```

3. **osbuild-selinux** - Required for SELinux-aware image builds on Fedora/RHEL hosts

```
sudo dnf install osbuild-selinux
```

4. **Sudo Access** - Required for loop device creation on Linux
 - The build script will automatically use **sudo** when needed
5. **Editor** - Examples use **vim**; **nano**, VS Code, or any editor works for YAML and kickstarts

2.2. Optional: Enable SSH for Remote Access

If you are connecting to the build system **remotely** (e.g., SSHing in from another machine to run builds), enable the SSH daemon:

```
systemctl enable sshd --now
```

This is not needed if you are working directly on the build machine.

2.3. System Requirements

Disk Space	At least 20 GB free (for packages, cache, and ISO)
Memory	4 GB minimum, 8 GB recommended
CPU	Multi-core processor recommended (build is CPU-intensive)
Internet	Required for downloading packages and container image

2.4. Supported Operating Systems

- ☐ Fedora Linux (39–44; newer releases track current Fedora Remix builder tags)
- ☐ RHEL/CentOS/Rocky/Alma Linux (8, 9)
- ☐ Ubuntu/Debian (with Podman installed)
- ☐ macOS (with Podman Desktop)

3. Quick Start (5 Steps)

3.1. Step 1: Clone the Repository

```
# Clone the Fedora Remix repository
git clone https://github.com/tmichett/Fedora_Remix.git

# Navigate to the project directory
cd Fedora_Remix
```

3.2. Step 2 (OPTIONAL - covered in Step 3): Container-specific settings in `config.yml`

The repository root `config.yml` holds `Container_Properties` used by `Build_Remix.sh` / `Verify_Build_Remix.sh`. You do **not** have to edit this file by hand: `./Update_Remix_Config.sh` (Step 3) prompts for `SSH_Key_Location`, `Fedora_Remix_Location`, `GitHub_Registry_Owner`, `Fedora_Version`, and `include_pxeboot_files` (PXE), writing values back to YAML safely.

When to edit here first: copy in a `config.yml` from another machine, or set paths/registry before running the script — those values become the **[bracket]** defaults at each prompt. You **may** still set `Fedora_Version` temporarily; Step 3 aligns `Fedora_Version`, `fedora_version` (`Setup/config.yml`), and PXE whenever you run it.

```
vim config.yml
```

Listing 1. Required Settings

```
Container_Properties:
  Fedora_Version: "43"                                # Prefer *Update_Remix_Config.sh*
  (Step 3) to keep both configs in sync
  SSH_Key_Location: "~/.ssh/github_id"                 # Your SSH key location
  Fedora_Remix_Location: "/home/travis/Remix_Builder" # Output directory
  GitHub_Registry_Owner: "tmichett"                   # Container registry owner
```

Table 1. Key Configuration Options

Option	Description
<code>Fedora_Version</code>	String form of the Fedora release used in the builder image tag (<code>ghcr.io/.../fedora-remix-builder:{version}</code>). Prefer setting it with <code>Update_Remix_Config.sh</code> so it matches <code>Setup/config.yml</code> .

Option	Description
<code>SSH_Key_Location</code>	Path to your SSH key (used for git operations in container)
<code>Fedora_Remix_Location</code>	Where the ISO and build artifacts will be saved
<code>GitHub_Registry_Owner</code>	Used in <code>ghcr.io/{owner}/fedora-remix-builder:{tag}</code> . If you pull the published <code>ghcr.io/tmichett/...</code> image, keep this <code>tmichett</code> (the script reminds you before this prompt).

Listing 2. Example Configuration

```
Container_Properties:
  Fedora_Version: "43"
  SSH_Key_Location: "~/.ssh/id_rsa"
  Fedora_Remix_Location: "/home/myuser/FedoraBuilds"
  GitHub_Registry_Owner: "tmichett"
```

3.3. Step 3: Paths, registry, Fedora release, and PXE (Update_Remix_Config.sh — recommended)

Instead of manually editing `Container_Properties` in `config.yml` and `fedora_version` / `include_pxeboot_files` in `Setup/config.yml`, run from the repository root:

```
chmod +x Update_Remix_Config.sh # once (if needed)
./Update_Remix_Config.sh
```

Prompts (in order) — Enter keeps the **[bracket]** default (from `config.yml` if present):

1. `SSH_Key_Location` — SSH key path for git operations in the container workflow
2. `Fedora_Remix_Location` — host directory for ISO output and build artifacts
3. `GitHub_Registry_Owner` — used to build `ghcr.io/{owner}/fedora-remix-builder:{tag}`; the script prints a reminder that if you use the **published** image under `ghcr.io/tmichett/...`, this value should stay `tmichett`
4. **Fedora release** — sets `Fedora_Version` and `fedora_version` together
5. **PXE/Linux boot images** — whether to download **PXE Linux** artifacts into the web prepare step (`include_pxeboot_files`). When **true**, container prepare stages `vmlinux` and `initrd.img` for optional **network (PXE/HTTP) boot** tooling on the Remix. If you answer **no**, PXE assets are skipped.



For some **older**, EOL, or **very new** Fedora versions, those images may be **unavailable** or **not** at the paths `Prepare_Web_Files.py` expects—the prepare or build can **fail**. If you are **not** using PXE/Linux network boot from this workflow, answer **no** (`include_pxeboot_files: false`).



The `fedora_version` in **Setup/config.yml** **MUST match** `Fedora_Version` in the root `config.yml`. **Update_Remix_Config.sh** updates both together to avoid drift.

Manual YAML (only if you did not run `Update_Remix_Config.sh`)

Edit **Setup/config.yml** so `fedora_version` matches `Fedora_Version`, and set `include_pxeboot_files` explicitly.

```
vim Setup/config.yml
```

```
fedora_version: 43
web_root: "/var/www/html"
include_pxeboot_files: false
```

3.4. Step 4: Verify Configuration (Recommended)

Run the verification script to check your configuration:

```
./Verify_Build_Remix.sh
```

What it checks

- `Fedora_Version` (`config.yml`) matches `fedora_version` (`Setup/config.yml`)
- Prints `include_pxeboot_files` in the remix summary panel (PXE payloads for web prepare)
- If `include_pxeboot_files` is missing from `Setup/config.yml`, prompts once and persists the answer (unless `REMIX_INCLUDE_PXEBOOT=true|false` is set in the environment)
- Resolves `ghcr.io/{owner}/fedora-remix-builder:{tag}`, confirms or pulls image
- Confirms before handing off to `./Build_Remix.sh` — passes `REMIX_INCLUDE_PXEBOOT` through so container prepare honors your PXE choice

Listing 3. Sample Output

```
┌────────────────────────────────────────────────────────────────────────────────┐
│ Configuration Summary                                                         │
├────────────────────────────────────────────────────────────────────────────────┤
│ Container Configuration (config.yml)                                         │
│   Fedora Version: 43                                                         │
│ Remix Configuration (Setup/config.yml)                                       │
│   Fedora Version: 43                                                         │
└────────────────────────────────────────────────────────────────────────────────┘
```

```
□ Versions match! Container and Remix both use Fedora 43
```

```
Do you want to proceed with the build? [y/N]: y
```

3.5. Step 5: Build the ISO

If you confirmed at the end of **Step 4**, the build starts automatically (`Verify_Build_Remix.sh` invokes `./Build_Remix.sh`).

Otherwise:

```
./Build_Remix.sh
```

3.5.1. Default detached mode / log streaming

Unless you pass `--attach` or set `REMIX_BUILD_ATTACH=1`, `./Build_Remix.sh` keeps the builder container detached and streams output by following `/tmp/entrypoint.log` in your current terminal (no separate `podman logs` pane required).

3.5.2. Stop container after ISO work

When the log follow ends:

```
podman stop remix-builder
```

Match `sudo` to how invoked on your machine (`sudo podman stop remix-builder`, etc.).

3.5.3. Attach-old-behavior toggle

Pass `--attach` (or `REMIX_BUILD_ATTACH=1`) only if you want the shell attached directly to the container PID 1 stream instead of the host-followed `/tmp/entrypoint.log`.

3.5.4. REMIX_INCLUDE_PXEBOOT (PXE payloads)

`Verify_Build_Remix.sh` exports `REMIX_INCLUDE_PXEBOOT` consistently with `include_pxeboot_files`. Driving builds directly:

```
REMIX_INCLUDE_PXEBOOT=false ./Build_Remix.sh
```

3.5.5. Typical entryptpoint milestones

1. systemd + hydration (patches bind-mounted from repo)
2. `dnf` installs (~15–25 minutes wall clock)
3. Kickstart `%post`, squashfs assemble, xorriso (~5–15 minutes)

Listing 4. ISO path (GNOME spin default kickstart)

```
/home/travis/Remix_Builder/FedoraRemix/FedoraRemix.iso
```

4. Optional Customization

4.1. Customizing Packages

To add or remove RPM packages from your remix:

Listing 5. Edit the package list

```
vim Setup/Kickstarts/FedoraRemixPackages.ks
```

Listing 6. Add packages

```
# Add your packages here
vim
tmux
htop
```

Listing 7. Remove packages

```
# Comment out or delete unwanted packages
# firefox # Removed
```

Package Categories

- Base system packages
- Desktop environment packages
- Development tools
- Multimedia applications
- Custom applications

4.2. Customizing Kickstart Files

For advanced customization of the installation and system configuration:

Listing 8. Main Kickstart Files

```
Setup/Kickstarts/
|—— FedoraRemix.ks           # Main kickstart (default)
|—— FedoraRemixCosmic.ks     # COSMIC desktop variant
|—— FedoraRemixKDE.ks        # KDE Plasma variant
|—— FedoraRemixLab.ks        # Lab/Development variant
|—— FedoraRemixPackages.ks   # Package list (edit this for packages)
```

```
└── FedoraRemixRepos.ks      # Repository configuration
```

Listing 9. Kickstart Snippets

```
Setup/Kickstarts/KickstartSnippets/  
└── create-ansible-user.ks    # Create ansible user  
└── customize-anaconda.ks    # Anaconda installer customization
```

What you can customize

- System packages and package groups
- User accounts and passwords
- Network configuration
- Firewall rules
- SELinux settings
- Post-installation scripts
- Boot loader configuration
- Partition layout
- Repository sources

Example: Adding a Custom User

Create a new snippet in `Setup/Kickstarts/KickstartSnippets/`:

```
vim Setup/Kickstarts/KickstartSnippets/create-myuser.ks
```

```
# Create custom user  
user --name=myuser --groups=wheel --password=encrypted_password --iscrypted
```

Then include it in your main kickstart:

```
vim Setup/Kickstarts/FedoraRemix.ks
```

```
%include Setup/Kickstarts/KickstartSnippets/create-myuser.ks
```

4.3. Customizing Look and Feel

Listing 10. Branding and Themes

```
Setup/files/  
├── boot/                # Boot splash screens  
│   ├── splash.png      # Boot menu background  
│   └── anaconda_header.png # Installer header  
├── FedoraRemixOrange.png # Desktop wallpaper  
└── Cursor.desktop      # Desktop shortcuts
```

Listing 11. GRUB Theme

```
Grub_Theme/Fedora_Remix_Theme/  
├── FedoraRemix.jpeg      # GRUB background options  
├── FedoraRemix2.jpeg  
└── ...
```

To customize

1. Replace image files with your own (keep same dimensions)
2. Update references in kickstart files if needed
3. Rebuild the ISO

5. Building Different Variants

The project includes several pre-configured variants:

5.1. Default Fedora Remix (GNOME)

```
./Verify_Build_Remix.sh  
# Select: 1) FedoraRemix
```

5.2. COSMIC Desktop Variant

```
./Build_Remix.sh --kickstart FedoraRemixCosmic
```

5.3. KDE Plasma Variant

```
./Build_Remix.sh --kickstart FedoraRemixKDE
```

5.4. Lab/Development Variant

```
./Build_Remix.sh --kickstart FedoraRemixLab
```

Each variant includes

- Different desktop environment
- Variant-specific packages
- Customized configurations
- Optimized for different use cases

6. Configuration File Reference

6.1. Main Configuration (**config.yml**)

Listing 12. Location

```
/home/travis/Github/Fedora_Remix/config.yml
```

Purpose

Controls the container build environment

Listing 13. Full Example

```
Container_Properties:
  # Fedora version for the container (must match Setup/config.yml)
  Fedora_Version: "43"

  # SSH key for git operations (optional but recommended)
  SSH_Key_Location: "~/.ssh/github_id"

  # Output directory for ISO and build artifacts
  Fedora_Remix_Location: "/home/travis/Remix_Builder"

  # GitHub Container Registry owner (for pulling container image)
  GitHub_Registry_Owner: "tmichett"

  # Note: Image_Name is auto-generated as:
  # ghcr.io/{GitHub_Registry_Owner}/fedora-remix-builder:{Fedora_Version}
```

6.2. Setup Configuration (**Setup/config.yml**)

Listing 14. Location

```
/home/travis/Github/Fedora_Remix/Setup/config.yml
```

Purpose

Controls the remix build process

Listing 15. Full Example

```
# Configuration file for Prepare_Web_Files.py

# Fedora boot files to download for PXE boot
```

```
fedora_boot_files:
  - "vmlinuz"
  - "initrd.img"

# Fedora version to use for downloading boot files
# MUST match Fedora_Version in main config.yml
fedora_version: 43

# Web root directory where files will be served
web_root: "/var/www/html"

# Download vmlinuz/initrd for PXE when true (see Update_Remix_Config.sh / Quick Start)
include_pxeboot_files: false
```

7. Troubleshooting

7.1. Version Mismatch Error

Listing 16. Problem

```
❏ Version mismatch detected!  
Container is configured for Fedora 43  
Remix is configured for Fedora 42
```

Solution

Run `./Update_Remix_Config.sh` so `Fedora_Version` and `fedora_version` match, or edit both files to the same release:

```
./Update_Remix_Config.sh  
# or  
vim config.yml  
vim Setup/config.yml
```

7.2. Container Image Not Found

Listing 17. Problem

```
❏ Container image not found locally
```

Solution 1: Let it pull automatically (slower)

- The image will be pulled from GitHub Container Registry
- Takes 5-15 minutes depending on connection

Listing 18. Solution 2: Build container locally (faster for repeated builds)

```
cd /home/travis/Github/RemixBuilder  
./build.sh
```

7.3. SELinux Relabeling Errors

Listing 19. Problem

```
setfiles: Could not set context for /usr/share/accountsservice: Invalid argument
```

```
Error creating Live CD : SELinux relabel failed.
```

Solution

This is fixed in the latest version. Ensure you have the patched `kickstart.py`:

```
ls -lh Setup/files/Fixes/kickstart.py
```

If missing, pull the latest changes:

```
git pull origin main
```

See [SELINUX_RELABEL_FIX.md](#) for details.

7.4. Build Fails After 15 Minutes

Listing 20. Problem

```
Error creating Live CD : Unable to unmount filesystem at /var/tmp/imgcreate-  
*/install_root/sys
```

Solution

This is fixed in the latest version. Ensure you have the patched `fs.py`:

```
ls -lh Setup/files/Fixes/fs.py
```

See [LINUX_BUILD_FIX.md](#) for details.

7.5. Insufficient Disk Space

Listing 21. Problem

```
Error: Not enough free space
```

Solution

- Ensure at least 20 GB free space
- Clean up old builds:

```
rm -rf /home/travis/Remix_Builder/FedoraRemix/*.iso
```

```
rm -rf /livecd-creator/package-cache/*
```

8. Build Output

8.1. Successful Build

Listing 22. Expected output

```
[[ Live CD created successfully!
  [ Total Build Time:          30m 46s (1846 seconds)
  [ Package Installation:      15m 42s (942 seconds)
  [ ISO File Creation:         15m 4s (904 seconds)
  [ Generated: FedoraRemix.iso (7.9G)
```

```
┌────────────────────────────────────────────────────────────────────────────────┐
│ BUILD COMPLETED SUCCESSFULLY! ──────────────────────────────────────────── │
└────────────────────────────────────────────────────────────────────────────────┘
```

8.2. Output Files

Listing 23. ISO Image

```
/home/travis/Remix_Builder/FedoraRemix/FedoraRemix.iso
```

Listing 24. Build Log

```
/home/travis/Remix_Builder/FedoraRemix/FedoraRemix-Build-MMDDYY-HHMM.log
```

Listing 25. Package Cache

```
/livecd-creator/package-cache/
```

9. Next Steps

9.1. Testing Your ISO

1. Verify the ISO:

```
ls -lh /home/travis/Remix_Builder/FedoraRemix/*.iso
```

2. Test in a VM:

- Use GNOME Boxes, VirtualBox, or virt-manager
- Boot from the ISO
- Test Live environment
- Test installation

3. Create Bootable USB:

```
sudo dd if=/home/travis/Remix_Builder/FedoraRemix/FedoraRemix.iso \
of=/dev/sdX bs=4M status=progress
```



Replace `/dev/sdX` with your USB device.

9.2. Distributing Your ISO

Options

- Upload to file sharing service
- Host on web server
- Distribute via USB drives
- Create torrent for large-scale distribution

10. Advanced Topics

10.1. Building for Different Fedora Versions

To build for a different Fedora version:

1. Run `./Update_Remix_Config.sh` and enter the new release (or update `Fedora_Version` and `fedora_version` together):

```
./Update_Remix_Config.sh
```

```
# or edit both files manually
# config.yml -> Fedora_Version: "44"
# Setup/config.yml -> fedora_version: 44
```

2. Verify the container image exists:

```
podman pull ghcr.io/tmichett/fedora-remix-builder:44
```

3. Build as normal

10.2. Automating Builds

Create a build script:

```
#!/bin/bash
# automated-build.sh

cd /home/travis/Github/Fedora_Remix

# Update to latest code
git pull

# Run build with auto-confirmation
echo "y" | ./Verify_Build_Remix.sh

# Copy ISO to distribution directory
cp /home/travis/Remix_Builder/FedoraRemix/*.iso /var/www/html/isos/
```

10.3. Building Multiple Variants

```
#!/bin/bash
# build-all-variants.sh

for variant in FedoraRemix FedoraRemixCosmic FedoraRemixKDE; do
    echo "Building $variant..."
    ./Build_Remix.sh --kickstart $variant
done
```

11. Additional Resources

11.1. Documentation

- **AsciiDoctor index:** [docs/README.adoc](#)
- **Physical/quickstart (ADOC/PDF):** [README_Physical.adoc](#)
- **Config helper:** [Update_Remix_Config.sh](#) — align `SSH_Key_Location`, `Fedora_Remix_Location`, `GitHub_Registry_Owner`, `Fedora_Version`, `fedora_version`, PXE (`include_pxeboot_files`)
- **Main README:** [README.md](#) - Project overview
- **Build Fixes:** [LINUX_BUILD_FIX.md](#) - Known issues and solutions
- **SELinux Fix:** [SELINUX_RELABEL_FIX.md](#) - SELinux relabeling fix details
- **Verification Script:** [VERIFY_BUILD_REMIX_USAGE.md](#) - Verification tool guide
- **Container Build:** [/home/travis/Github/RemixBuilder/README.md](#) - Container documentation

11.2. Repository Links

- **Fedora Remix:** https://github.com/tmichett/Fedora_Remix
- **Remix Builder Container:** <https://github.com/tmichett/RemixBuilder>
- **Container Registry:** <https://ghcr.io/tmichett/fedora-remix-builder>

11.3. Getting Help

1. Check the documentation files
2. Review build logs for errors
3. Check GitHub Issues
4. Consult Fedora documentation for kickstart syntax

12. Summary

Minimum steps to build

1. `git clone …/Fedora_Remix.git && cd Fedora_Remix`
2. `./Update_Remix_Config.sh` — `SSH_Key_Location`, `Fedora_Remix_Location`, `GitHub_Registry_Owner` (keep `tmichett` if using `ghcr.io/tmichett/fedora-remix-builder`), `Fedora_Version`, PXE (**no** unless serving PXE); optional pre-edit `config.yml` for defaults
3. `./Verify_Build_Remix.sh` (or `./Build_Remix.sh`)
4. `podman stop remix-builder` when finished inspecting logs
5. ISO beside `${Fedora_Remix_Location}/FedoraRemix/` (`/home/travis/Remix_Builder/…` in examples)

Optional Customization

- Edit `Setup/Kickstarts/FedoraRemixPackages.ks` - Add/remove packages
- Edit kickstart files in `Setup/Kickstarts/` - Advanced customization
- Replace branding files in `Setup/files/` - Custom look and feel

That's it! You now have a custom Fedora Remix ISO ready to use.

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13. Fedora Remix Creation Quickstart (Physical or Virtual)

This quickstart will provide a quick method to download a pre-built Fedora Remix as well as creating your own Fedora Remix Building system. The https://github.com/tmichett/Fedora_Remix has been created with Python automation scripts that will setup a Fedora Remix Building system using the Fedora Remix installed from the liveCD provided (https://drive.google.com/drive/folders/1UAT07AJrTdMk3ke_QE6S6vz-qn6qGtR). The requirements are relatively simply to build and the system has Fedora Remix customization scripts and self-launching menu to assist with the initial setup. The quickstart and documentation are also available at the following link: https://tmichett.github.io/Fedora_Remix/.

For a Markdown walkthrough that tracks the same flow as **Build_Remix_Physical.sh**, see [Quickstart_Physical.md](#) in the repository.

14. Fedora release and PXE (Update_Remix_Config.sh)

After you clone the repository, run **Update_Remix_Config.sh** from the repository root (recommended) instead of manually editing **config.yml** and **Setup/config.yml**.

```
chmod +x Update_Remix_Config.sh
./Update_Remix_Config.sh
```

It prompts for **SSH_Key_Location**, **Fedora_Remix_Location**, **GitHub_Registry_Owner**, the Fedora release, and whether to download **PXE Linux** boot images for the web tree. It updates **config.yml** (**Container_Properties**), **Setup/config.yml** **fedora_version**, and **include_pxeboot_files**. If you use the published container image at ghcr.io/tmichett/fedora-remix-builder, keep **GitHub_Registry_Owner** set to **tmichett**. When **include_pxeboot_files** is **true**, **Prepare_Web_Files.py** fetches **vmlinux** and **initrd.img** so you can optionally use the build host (HTTPd) as part of a **PXE / network-install** layout for your Fedora Remix.



For some **older** releases, EOL composes, or **very new** Fedora versions, those kernel and initrd artifacts may be **missing** or not at the expected mirror paths; including PXE assets can make **Prepare_Web_Files.py** or later steps **fail**. If you are **not** planning to serve PXE/network-boot content from this system, answer **no** and keep **include_pxeboot_files: false**.

Building your own Fedora Remix - Overview

1. Download the Fedora Remix ISO and install locally to hard drive
 - a. Setup a local user and ensure it is in the SUDOERS file
 - b. Run some the Fedora Remix customize scripts to create SSH keys and SUDO with no password
2. Clone https://github.com/tmichett/Fedora_Remix
3. Run **Update_Remix_Config.sh** from the repository root (recommended) to align **Container_Properties** (including **Fedora_Version** and registry fields), **fedora_version**, and PXE-related **include_pxeboot_files**; or set those values by hand in **config.yml** and **Setup/config.yml**
4. Either run **Build_Remix_Physical.sh** from the repository root (recommended), or run the Python automation scripts manually with **sudo** from the **Setup** directory in this order:
 - a. **Prepare_Fedora_Remix_Build.py** (creates **/livecd-creator/FedoraRemix** and copies assets)
 - b. **Prepare_Web_Files.py** (HTTPd, web assets, and imgcreate **kickstart.py / fs.py** fixes under **/var/www/html/**)
5. Optionally modify the kickstart files in **/livecd-creator/FedoraRemix** after the prepare step (or edit under **Setup/Kickstarts/** before building)

- a. **FedoraRemix.ks** - Control various setup components and run commands
 - b. **FedoraRemixPackages.ks** - Control RPM packages installed in your Remix
6. Run the enhanced build: **Enhanced_Remix_Build_Script.sh** in `/livecd-creator/FedoraRemix` (invoked for you by **Build_Remix_Physical.sh**). **Remix_Build_Script.sh** remains available as a legacy wrapper



Error Regarding kickstart.py

With newer Fedora releases, there is an error with the **kickstart.py** package and URLGrabber. Located on the "FedoraRemixTools" application, I've included a button to "FIX" the issue with the **kickstart.py** package. After completing the fix, you will no longer get the issue.

Listing 26. Python kickstart.py Errors

```
Traceback (most recent call last):
  File "/usr/sbin/livecd-creator", line 265, in <module>
    sys.exit(main())

... OUTPUT OMITTED ...
```

14.1. Quickstart

This quickstart provides detailed information on how to download, install, and create your custom Fedora Remix. It depends first on having the Fedora Remix installed locally and access to the Github repository and playbooks contained within the repository.

1. Download the "Latest" FedoraRemix LiveISO and create a bootable USB disk.
 - https://drive.google.com/drive/folders/1UAT07AJIrTdMk3ke_QE6S6vz-qn6qGtR
 - Use Balena Etcher or Fedora Media Writer to create the bootable USB device for installation on a physical machine.
 - Attach the ISO image to your virtual machine and select it as the boot device so it can be installed
2. Insert the disk into a computer or connect to the virtual machine and select

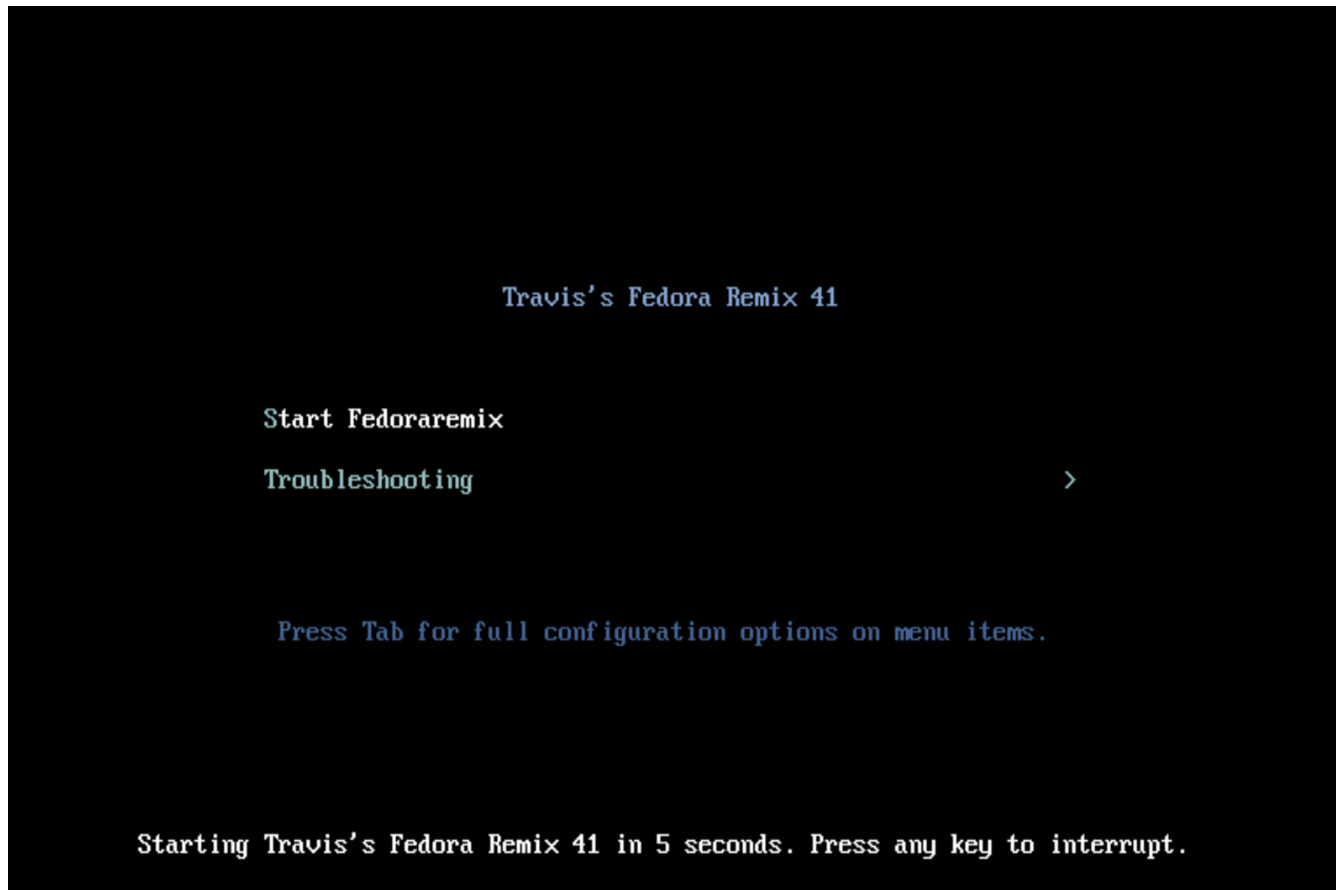


Figure 1. GRUB Boot Menu

3. Select "Start Fedoraremix" and hit Enter
4. Once booted to the main screen, click on the icon in the top left beside of "Apps" to bring up the application dock. Select the icon from the dock

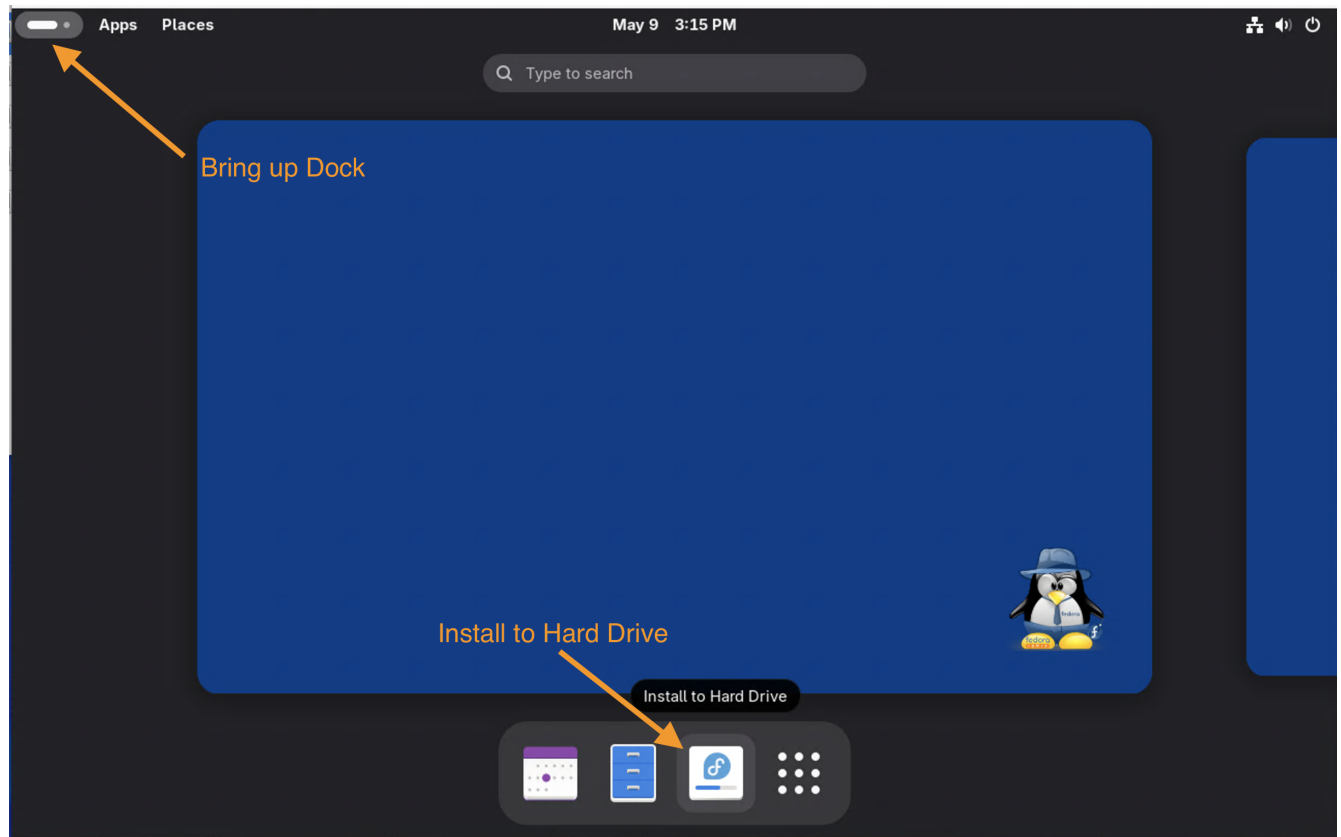


Figure 2. Fedora Remix LiveCD Desktop

5. The Anaconda graphical installer will appear. Select your desired language and optionally select the keyboard layout.

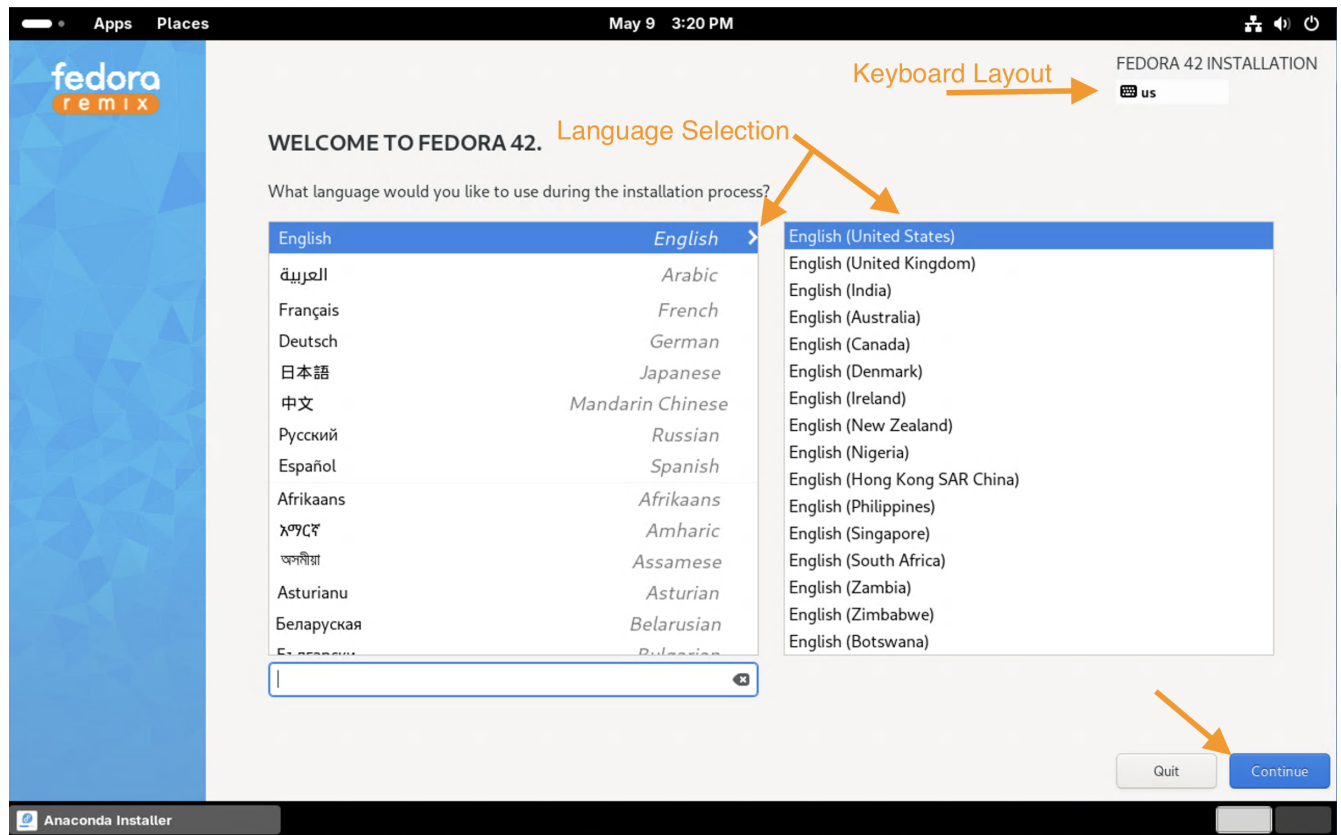


Figure 3. Anaconda Installer - Language and Keyboard

6. Complete the required and optional installation components.

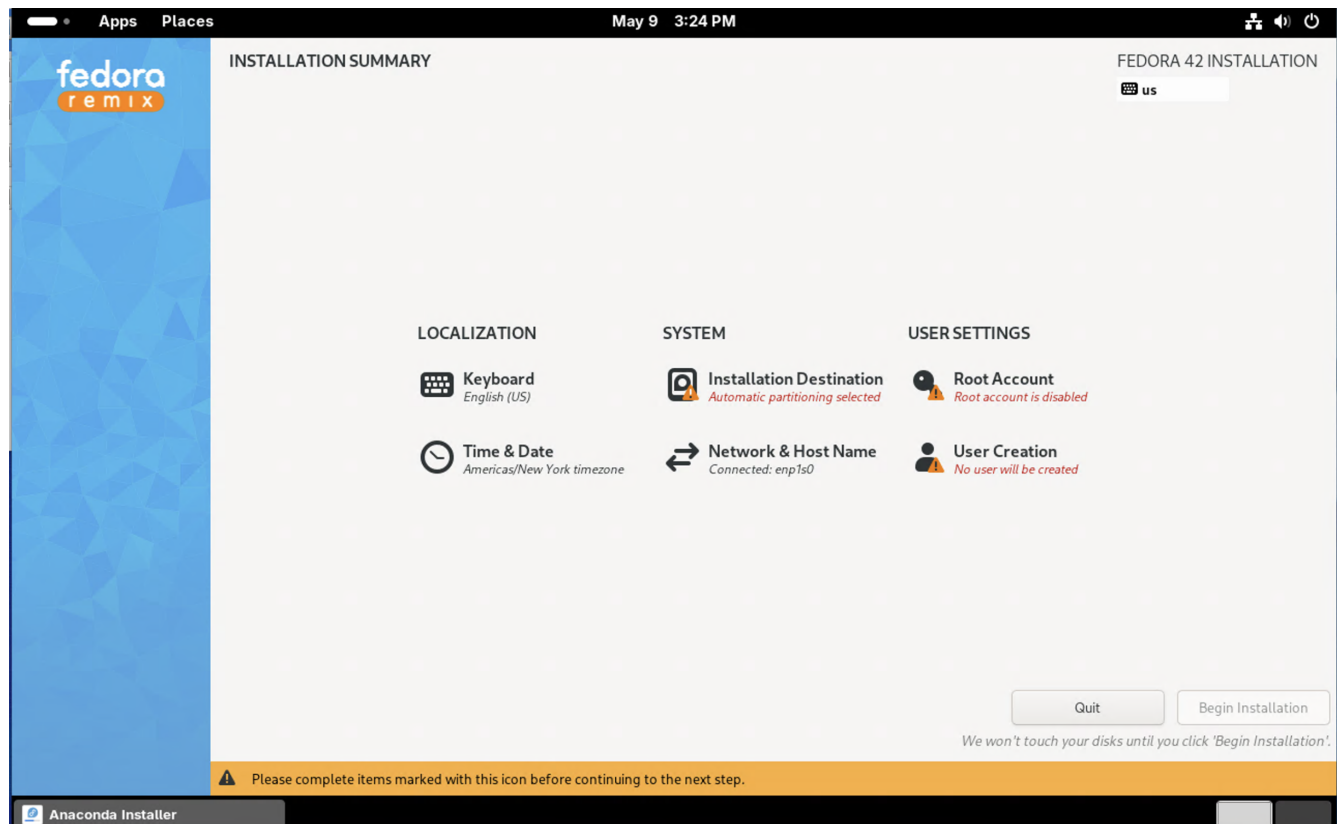


Figure 4. Anaconda Installer - Installation Summary

- a. Select "Installation Destination" and choose an appropriate installation device.

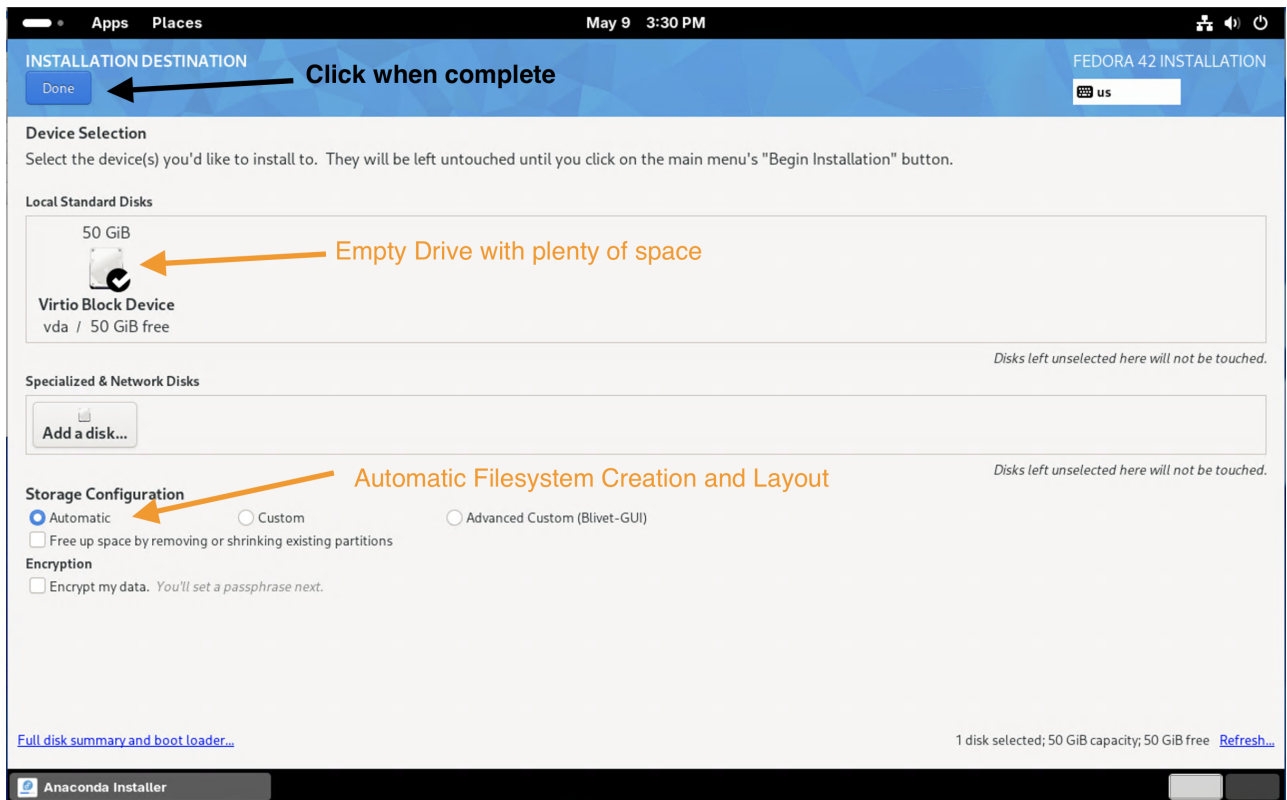


Figure 5. Setting Installation Location

- b. Create a user account by clicking "User Creation". Complete the required information and then click done.

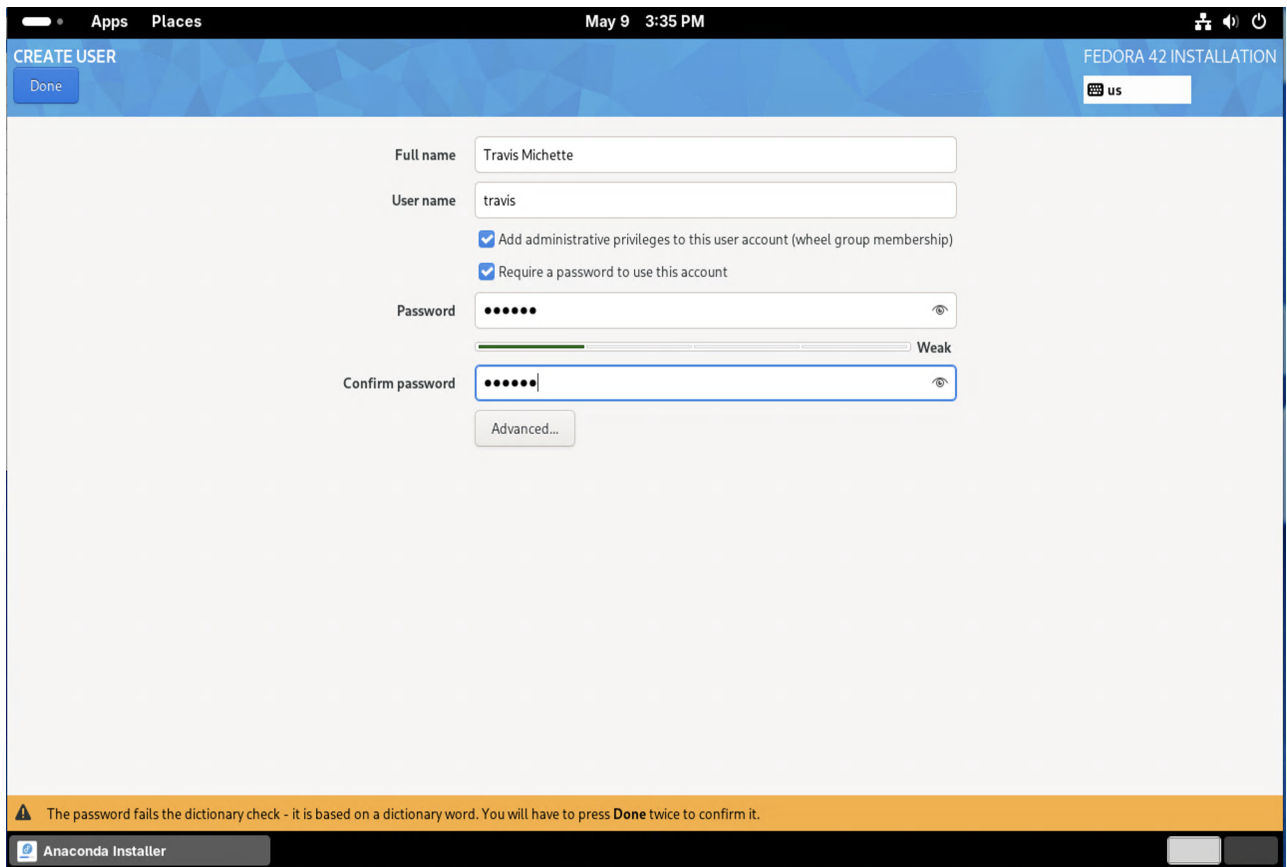


Figure 6. User Creation



Root Account

After you have specified the installation location and an initial user, you may proceed with the installation. Creating and enabling the **root** user is optional and not required. If you create the root user, likewise, you are not required to create a regular user account.



Setup Complete

After the installation is completed, you will need to reboot your system and perform package updates and any other needed tasks.

7. Clone the Fedora Remix repository

```
travis@localhost-fedoraremix-live [~] | (05/09/25 @ 15:55)
$ git clone https://github.com/tmichett/Fedora_Remix.git
Cloning into 'Fedora_Remix'...
remote: Enumerating objects: 1949, done

... OUTPUT OMITTED ...

Resolving deltas: 100% (1042/1042), done.
```

8. **Recommended:** from the repository root, run **Build_Remix_Physical.sh**. It prompts for the Fedora release (written to **Setup/config.yml** as **fedora_version**; or pre-set Fedora and PXE preferences with **Update_Remix_Config.sh**), offers a kickstart choice, then runs **Prepare_Fedora_Remix_Build.py**, **Prepare_Web_Files.py**, and **Enhanced_Remix_Build_Script.sh** in **/livecd-creator/FedoraRemix**. Use **./Build_Remix_Physical.sh -h** for options (**-v**, **-k**, **-l**).

Listing 27. Run Build_Remix_Physical.sh (native / physical build driver)

```
travis@localhost-fedoraremix-live [~/Fedora_Remix] | main ✓ |
$ chmod +x Build_Remix_Physical.sh
$ ./Build_Remix_Physical.sh

... prompts for version, kickstart, confirmation; then prepare + enhanced build ...
```

9. **Alternative — manual prepare:** run the Python scripts from the **Setup** directory as **root**, in this order (same as **Build_Remix_Physical.sh**):

Listing 28. Create /livecd-creator/FedoraRemix assets

```
travis@localhost-fedoraremix-live [~/Fedora_Remix/Setup] | (05/09/25 @ 15:57) | main
✓ |
$ sudo python3 Prepare_Fedora_Remix_Build.py

... OUTPUT OMITTED ...

Copying ../Remix_Build_Script.sh to /livecd-creator/FedoraRemix/Remix_Build_Script.sh
Setup complete!
```

Listing 29. Create httpd server and repository assets

```
travis@localhost-fedoraremix-live [~/Fedora_Remix/Setup] | (05/09/25 @ 15:59) | main
✓ |
$ sudo python3 Prepare_Web_Files.py
Installing packages: httpd
Running command: dnf install -y httpd

... OUTPUT OMITTED ...

Setup complete!
```

10. Perform system and package customizations by modifying the **FedoraRemix.ks** kickstart file and the **FedoraRemixPackages.ks** file (under **/livecd-creator/FedoraRemix** after the prepare step). The default **Setup/config.yml** **fedora_version** is updated for you if you use **Build_Remix_Physical.sh**; otherwise use **Update_Remix_Config.sh** or set **fedora_version** (and PXE settings) before or between prepare steps.

- a. The setup scripts create the `/livecd-creator/FedoraRemix` directory. That is where you edit kickstarts and where the ISO (e.g. **FedoraRemix.iso**) is written after a successful build.

Listing 30. FedoraRemix.ks

```
... OUTPUT OMITTED ...

## Add Fedora Dynamic MotD Script
cd /usr/bin
wget http://localhost/files/fedora-dynamic-motd.sh
chmod +x /usr/bin/fedora-dynamic-motd.sh
echo /usr/bin/fedora-dynamic-motd.sh >> /etc/profile

## Customize BASH Prompts and Shell
mkdir /opt/bash
cd /opt/bash
wget http://localhost/files/bashrc.append
## Install Gitprompt
git clone https://github.com/tmichett/bash-git-prompt.git /opt/bash-git-prompt
--depth=1

... OUTPUT OMITTED ...
```

Listing 31. FedoraPackages.ks

```
... OUTPUT OMITTED ...

## Image Editing and Manipulation
inkscape
gimp
krita
netpbm-progs
scribus

## Video Editing and Manipulation
kdenlive

## Container Tools
buildah
skopeo
podman-machine

... OUTPUT OMITTED ...
```

11. Execute **Enhanced_Remix_Build_Script.sh** in `/livecd-creator/FedoraRemix` to start the build (or

rely on the last step of **Build_Remix_Physical.sh**). For a non-default kickstart, set **REMIX_KICKSTART** (e.g. `sudo env REMIX_KICKSTART=FedoraRemixCosmic ./Enhanced_Remix_Build_Script.sh`). Depending on system and network speed and kickstart changes, the process can take on the order of an hour.

Listing 32. Launch the enhanced build script

```
travis@localhost-fedoraremix-live [/lived-creator/FedoraRemix] | (05/09/25 @ 16:09)
$ time sudo ./Enhanced_Remix_Build_Script.sh
/usr/lib/python3.13/site-packages/pykickstart/commands/partition.py:461:
KickstartParseWarning: A partition with the mountpoint / has already been defined.
  warnings.warn(_("A partition with the mountpoint %s has already been defined.") %
pd.mountpoint, KickstartParseWarning)
fedora                19 MB/s | 35 MB      00:01
updates              3.8 MB/s | 5.8 MB      00:01
google-chrome         7.5 kB/s | 1.9 kB      00:00

... OUTPUT OMITTED ...

Pass 4: Checking reference counts
Pass 5: Checking group summary information
_FedoraRemix: 371986/1324512 files (0.2% non-contiguous), 4248977/5294080 blocks

e2fsck 1.47.1 (20-May-2024)

0

real    34m41.254s
user    143m42.195s
sys     5m11.054s
```

Error Regarding kickstart.py

With newer Fedora releases, there is an error with the **kickstart.py** package and URLGrabber. Located on the "FedoraRemixTools" application, I've included a button to "FIX" the issue with the **kickstart.py** package. After completing the fix, you will no longer get the issue.



Listing 33. Python kickstart.py Errors

```
Traceback (most recent call last):
  File "/usr/sbin/lived-creator", line 265, in <module>
    sys.exit(main())
    ~~~~^^
  File "/usr/sbin/lived-creator", line 195, in main
    ks = imgcreate.read_kickstart(options.kscfg)
  File "/usr/lib/python3.13/site-packages/imgcreate/kickstart.py", line
```

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Appendix A: Fedora Remix

This Fedora Remix has been built and assembled from many projects and continues to evolve and become simplified. The various Github repositories are meant to allow easier updates to individual Fedora Remix components and to allow certain components to be used more in a utility fashion where they can be used without impacting the Fedora Remix project.

A.1. Fedora Remix Application Repositories

This project was initially 100% part of the Fedora Remix repository using various BASH scripts, Ansible Playbooks, Python tools, and various applications like YAD (Yet Another Dialog). As the project grew, some of the components have been separated out into their own repositories for use in the creation of the Fedora Remix.

- **Log Viewer** - (https://github.com/tmichett/log_viewer)

The Log Viewer was developed to look at and analyze the output of the kickstart file. During development and updates to the Fedora Remix build process as part of Fedora 41/42, the kickstart and build files became extremely large and it was difficult to identify errors and successes within a huge file, so the Log Viewer was created to look for keywords in that file and assist with the building and troubleshooting process. Using Github actions and workflows, this builds the Python application as a Linux executable with all dependencies as well as produces an RPM for easy installation in Fedora Remix.

- **PXE Server** - (<https://github.com/tmichett/PXEServer>)

The PXEServer repository was created to separate just the PXE-boot components from Fedora Remix. This may later be replaced as it will be a modular component and it was designed to allow a "lab" or "classroom-style" deployment where Fedora Remix would be installed to a single system and the PXE server could provide the Fedora Remix Live image across the network via PXE without ever installing anything to the remaining computers. This repository contains the items and automation for configuring PXE boot services and identifying the components.

- **Fedora Remix Customize** - (<https://github.com/tmichett/FedoraRemixCustomize>)

The Fedora Remix Customize repository contains some of the user-based customizations. It was created as a separate repository for tracking playbooks and scripts that would further customize the environment. Eventually, this will be the location of many more customization utilities as the build and deployment process gets more streamlined.

- **Fedora Remix Tools** - (https://github.com/tmichett/Fedora_Remix_Tools)

The Fedora Remix Tools repository brings a new take to the **Fedora Remix Apps** which was the first set of tools and customizations. The Fedora Remix Apps used YAD and was a set of BASH scripts and playbooks located in the main Fedora Remix repository in the YAD folder

(https://github.com/tmichett/Fedora_Remix/tree/main/YAD). The implementation and location of YAD made things more difficult to test and maintain as well as add new features. With the release of Fedora 42, a newer Python application has been developed. This application has a dynamic menu system that can be controlled by changes in a configuration file (config.yml) that allows simple text changes to change and modify the application look and feel. The newer Fedora Remix Tools application also provides a *command output window* which can be detached and looked at so a user knows if there were errors or issues running any of the customizations. The Fedora Remix Tools repository also uses Github actions and workflows to build an RPM which can be easily installed on the Fedora Remix system.

- **OhMyBash** - (<https://github.com/tmichett/oh-my-bash>)

The **OhMyBash** repository was forked from the main repository but customized for Fedora Remix. This repository is used to install OhMyBash as part of the Fedora Remix built process and perform the various customizations to the terminal and BASH shell including configuration and creation of a custom theme developed for the Fedora Remix (<https://github.com/tmichett/oh-my-bash/tree/master/themes/tm-fedora-remix>).

A.2. Fedora Remix Extensions and Additional Projects

There are two additional projects inspired by the Fedora Remix work which will be coming soon to Fedora Remix or stand slightly apart from Fedora Remix. The idea is to look at making a Live boot USB more interactive and not waste an entire thumb drive as well as experiment with PXE and other distribution methods including BootC container images.

- **Fedora BootC** - (https://github.com/tmichett/Fedora_BootC)

The Fedora BootC repository is in initial testing and development stages and is meant to be picked up later in 2025 to extend the customizations and create an image-based installer that can be used with the Fedora Remix PXE process that will allow a kickstart file to provision systems in lab or classroom-style environments rather than just a live image. This repository will also likely extend into research and initial starting point for creating a WSL image of Fedora Remix to run on Windows systems.

- **Fedora Remix Ventoy** - (https://github.com/tmichett/Fedora_Remix_Ventoy)

The Ventoy application has been around and is available for Windows and Linux allowing users to build USB drives with multiple ISOs creating a multi-boot USB. Ventoy can also be customized with many components, but it must be installed and users must know how to use Ventoy. The Fedora Remix Ventoy repository has been created to allow an RPM installer to bake in Ventoy into the Fedora Remix Live image. It also has a Python GUI application that is meant to bring up some of the Ventoy customization components without needing to know all the commands, procedures, and processes in an attempt to make things more accessible.